

Echoes of Information Flow: Tracing Volatility and Heavy-tailed Distribution in Cryptocurrency

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Abstract

We examine the role of information flow in shaping volatility and tail risk of cryptocurrency returns. Using a stable-GJR-GARCH model with trading activity as a subordinator, we find that information flow increases conditional volatility while reducing—but not eliminating—heavy tails. Effects are stronger for more liquid cryptos. *Trading days* log-returns remain non-Gaussian even after correcting for information-induced heteroskedasticity within *calendar days*, challenging Gaussian assumptions in risk and asset-pricing models.

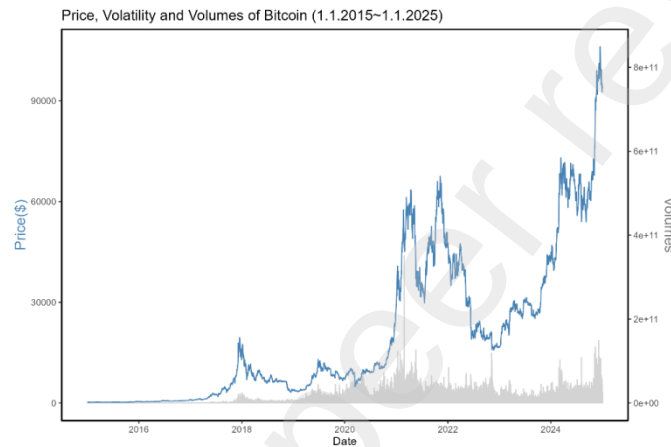
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1. Introduction

Cryptocurrencies entered a new era after the U.S. establish the Strategic Crypto Reserve and pass the Stablecoin Act in 2025. Being a reserve asset, the substantial volatility of cryptos (see Figure 1) may challenge financial stability. Since cryptos' return distribution displays heavy tails, conditional heteroskedasticity and volatility clustering, the probability of extreme losses and persistent fluctuation will be higher compared to the Gaussian distribution assumption (Fry, 2018; Malek et al., 2023). These characteristics highlight the importance of understanding and monitoring extreme risks (Bakó and Sándor, 2025). However, research gaps exist in providing theoretical ground to explain such



behavior.

Fig. 1. Price, Volatility and Volumes of Bitcoin

While the crypto pricing models developed by Liu et al. (2022) and Cong et al. (2021) have the potential to resolve the “bubble or asset” debate between Paul Krugman (2018) and Jean Tirole (2024), their Gaussian assumption is inconsistent with empirical evidence. Rietz (1988), Julliard and Ghosh (2012) and Fu et al. (2024) find that the equity-premium puzzle (Mehra and Prescott, 1985) is related to rare events and fat-tailed distribution. The shape of return distribution is influenced by information affecting the intrinsic value and how that information flows to the market (Damodaran, 1985). Whether arbitrage profits can be made in cryptocurrency markets is affected by information flow and its impact on prices, i.e., the statistical behavior of the price changes¹. The subordinated distribution

¹ Fama's (1965b, 1970) Efficient Market Hypothesis (EMH) states that asset prices fully and instantly reflect all available information, hence no one can earn arbitrage profits. On the other hand, Capital Asset Pricing model (CAPM) and the Option Pricing model (OPM) emphasize the importance of diversification and hedging mechanisms in managing risk. For cryptocurrency, Urquhart (2016) find that crypto market is not

(Clark, 1973) showed that price changes over “*calendar time*” reflect an accumulation of new information during a particular “*trading time*”, that is similar to the “operational” time scale of Stock (1988) and Diebold (1986). Due to the randomness of information arrival, the unconditional return distribution presents fat-tails within “calendar time”. It is a subordinated stochastic process with a mixture of normal distributions, in which information flow is the directing process. After adjusting for the random arrival of information, the logarithm of returns is suggested to follow *normal* distribution (hereafter log-normal) by Clark (1973), while *non-normal* stable distribution (hereafter log-stable) is recommended by Mandelbrot and Taylor (1967), Granger and Morgenstern (1970) and Mandelbrot (1997). Conflicting results remain in terms of the return and volatility patterns accounting for information flow.

Clark’s (1973) subordination process provides theoretical foundation for the generalized autoregressive conditional heteroskedasticity (GARCH, Bollerslev, 1986) model. Their relations are illustrated in Appendix A. Lamoureux and Lastrapes (1990, 1994) employ GARCH model to explain the time-varying relationships between information flow and volatility of stocks. They show that the persistence of volatility can be captured by trading volume², which serves as a proxy for information flow and acts as a subordinator, creating fat tails and volatility clustering as the ARCH (Engle, 1982) and GARCH models. In other words, autocorrelation in the time-varying rate of information arrival causes volatility dependency. Previous studies using GARCH and subordinated distribution with the Gaussian assumption cannot capture the extreme impact of information on volatility, *which could be depicted by stable distribution*. Additionally, stable distribution meets the Generalized Central Limit Theorem that other popular distributions do not satisfy (see Appendix C.2). Previous GARCH family models therefore should be challenged for examining crypto volatility.

This study investigates whether the unconditional fat-tailed return distribution is due to volatility persistence created by random arrival of information. We use a “composite” of

efficient, while Wei (2018) and Fry (2018) argue that liquidity must be considered. Whether cryptocurrency markets are efficient or not remains a controversial issue.

² Lamoureux and Lastrapes (1994) relax the assumption of exogenous volume and find that accounting for serial dependence in the information-arrival process does not eliminate GARCH persistence in variance.

stable distribution and GJR-GARCH model (see Table 1) to examine how information asymmetrically influences the extreme changes of cryptocurrencies' return and volatility.

Table 1. The Research Design of “Composite” Approach

Model/ Possible Causes	Gaussian	Non-Gaussian
Homoscedastic model	Benchmark AR models	Stable Paretian: Mandelbrot (1963), Fama (1965)
Conditional Heteroskedastic model	ARCH, GARCH, EGARCH, GJR-GARCH etc.: Engle (1982), Engle and Bollerslev (1986), Glosten et al. (1993)	Stable GARCH: Mittnik et al. (2002), Broda et al. (2013)
Channel: Information Arrival (trading volumes)	Subordinated distribution: calendar days V.S. trading days Clark (1973), Stock (1988), Lamoureux and Lastrapes (1990, 1994).	Log-stable: Mandelbrot and Taylor (1967), Granger and Morgenstern (1970) <i>*Current study: Stable-GJR-GARCH with trading volumes.</i>

This study makes several contributions: 1) It explains the role of information as a subordinator shaping the fat-tailed distribution of cryptocurrency. The stable-GJR-GARCH model captures the extreme impact of information on volatility. 2) It shows that “tails” of cryptos' volatility-adjusted distribution remain significantly “fatter” than Gaussian distribution. It challenges Gaussian-based risk and asset-pricing models in crypto markets. Stable distribution should be reconsidered because it provides more relevant and useful information for investment decisions.

2. Data and Methodology

2.1 Data

Our sample comprises all 248 cryptocurrencies in the market with complete time series from 1 January 2019 to 1 January 2025 covering the COVID-19 market wipeout. The closing prices, trading volumes, and market capitalization at daily intervals are retrieved from CoinMarketCap, consisting of 1,635,930 observations. Given that higher trading volumes indicate more analyst coverage and larger information flow (Damodaran, 1985), we sort cryptos into five quintiles based on daily trading volume to compare information's influence at different levels. Five value-weighted portfolios are constructed.

2.2 Stable distribution

The characteristic function $X \sim S(\alpha, \beta, \gamma, \delta)$ of stable distribution proposed by Mandelbrot (1963) and Fama (1965a) can be described as (Nolan, 2014, So, 1978a, 1987b):

$$E[\exp(iuX)] = \begin{cases} \exp(-\gamma^\alpha |u|^\alpha [1 - i\beta(\tan \frac{\pi\alpha}{2})(\text{sign}(u))] + i\delta u) & \alpha \neq 1 \\ \exp(-\gamma |u| [1 + i\beta \frac{2}{\pi}(\text{sign}(u)) \times \log(|u|)] + i\delta u) & \alpha = 1 \end{cases} \quad (1)$$

where $i = \sqrt{-1}$ is the imagine number. $\alpha \in (0, 2]$ denotes the characteristic exponent characterizing heaviness of the tails. When $\alpha = 2$, it corresponds to Gaussian distribution. The smaller the α , the heavier the tails. $\beta \in [-1, 1]$ describes skewness. α and β are the shape parameters depicting *long-term tail risk*. $\gamma > 0$ is the scale parameter measuring dispersion (*short-term volatility*)³. $\delta \in \mathbb{R}$ is the location parameter.

2.3 Stable-GJR-GARCH

The return in *calendar time* is expressed as (Glosten et al., 1993):

$$r_t = \mu + \varepsilon_t \quad (2)$$

Where μ is the mean return, $\varepsilon_t \sim S(\alpha, \beta, \gamma, \delta)$ is the stable distributed innovations. The conditional variance h_t is described as (Mittnik et al., 2002)⁴:

$$h_t^\eta = \varpi + \sum_{i=1}^q \lambda_i |\varepsilon_{i,t-i}|^\eta + \sum_{i=1}^o \varphi_i I_{t-i} |\varepsilon_{i,t-i}|^\eta + \sum_{j=1}^p \theta_j h_{j,t-j}^\eta \quad (3)$$

Where $\varpi > 0$ represents the baseline volatility, $I_{t-i} = 1$ if $\varepsilon_{i,t-i} < 0$ (negative news) and 0 otherwise. Parameter λ_i capture the ARCH effect (short-term volatility spikes), θ_j measures the GARCH effect (intermediate volatility persistence), φ_i captures the leverage effect. Coefficients λ_i ($i=0, \dots, p$), φ_i ($i=0, \dots, o$) and θ_j ($j=1, \dots, q$) are all non-negative. The power parameter, η , satisfies $0 < \eta < \alpha$.

When trading volume is added to explain the volatility as in Lamoureux and Lastrapes (1990, 1994), equation (3) takes the form as:

$$h_t^\eta = \varpi + \sum_{i=1}^q \lambda_i |\varepsilon_{i,t-i}|^\eta + \sum_{i=1}^o \varphi_i I_{t-i} |\varepsilon_{i,t-i}|^\eta + \sum_{j=1}^p \theta_j h_{j,t-j}^\eta + \xi v_{t-1} \quad (4)$$

³ Luo (2025) finds that volatility primarily drives the tail exponent. Trading volume significantly affects the scale parameter of extremely large returns in stocks, while its relations with the tail index remains unclear.

⁴ Also see Francq and Meintanis (2016), Liu and Brorsen (1995), Paolella (2016).

Where v_{t-1} denotes the log-change of volume, ξ represents volume's impact.

3. Empirical results

Table 2 shows the descriptive statistics and stable distribution estimates of crypto portfolios. It documents the heavy tails of unconditional return distribution in calendar days (kurtosis > 3 and $\alpha < 2$). As volumes become lower from Portfolio 1 to 5, the size (market capitalization) of portfolios decreases and tails tend to be heavier. It indicates smaller cryptos have higher probability of realizing extreme returns than the larger cryptos. The findings highlight the need for heavy-tailed distribution in modelling crypto returns and innovations.

Table 2. Descriptive Statistics of Crypto Portfolios

	1	2	3	4	5
	HIGH				LOW
Min (r_t)	-0.478	-0.554	-0.512	-0.496	-0.547
Mean (r_t)	0.002	0.002	0.003	0.006	0.012
Median (r_t)	0.002	0.004	0.005	0.004	0.006
Max (r_t)	0.169	0.284	0.245	0.564	1.558
Std (r_t)	0.036	0.042	0.039	0.052	0.083
Skew (r_t)	-1.391	-1.857	-1.558	0.831	7.424
Kurt (r_t)	21.146	23.895	23.040	22.300	121.282
$\alpha(r_t)$	1.543	1.631	1.609	1.460	1.383
$\beta(r_t)$	-0.047	-0.322	-0.172	0.040	0.179
$\gamma(r_t)$	0.017	0.021	0.019	0.021	0.022
$\delta(r_t)$	0.002	0.006	0.006	0.004	0.005
E ($ r_t $)	0.024	0.028	0.026	0.032	0.038
E (VOL_t)	22,626,232,738	15,055,346	2,871,718	341,958	66,682
E (CAP_t)	431,027,491,537	503,872,278	260,532,461	64,081,442	22,755,897
SW (r_t)	0.898	0.878	0.881	0.821	0.542

The portfolios are ranked in descending order based on daily trading volumes. r_t is the logarithm of portfolio's daily returns. E($|r_t|$) stands for the average absolute value of daily return. E(VOL_t) is the average daily trading volume. E(CAP_t) is the average daily market capitalization. SW (r_t) denotes the Shapiro-Wilk test statistics of crypto portfolio daily log-returns.

Table 3 provides the parameter estimates of GJR-GARCH(1,1) model under Gaussian assumption. It shows all portfolios exhibit significant ARCH and GARCH effects in both calendar time and trading time. The information flow, proxied by the log-change of volume, have significant and positive influence on volatility among all cryptos. The arrival of

information (ξ) displays larger impacts on larger size cryptos. The leverage effects become significant after accounting for information flow. In Table 3(A), the sum of coefficients ($\lambda + \theta + 1/2\gamma=0.999$) in Portfolios 4 & 5 is close to 1 as in Integrated-GARCH (Engle and Bollerslev, 1986), which has infinite variance and is non-stationary⁵. In Table 3(B), the sum of coefficients becomes larger than 1 in Portfolios 1, 4 & 5. The findings show significant impact of information flow, yet Gaussian distribution is inappropriate for modelling the process.

Table 3. Parameter Estimates of Normal-GJR-GARCH(1,1) Model

	1		2		3		4		5
Panel A. Without information arrival (<i>In calendar time</i>)									
ω	0.00011	*	0.00014	*	0.00005		0.00008		0.00017
	(1.86)		(1.68)		(0.75)		(1.6)		(1.06)
λ	0.058	**	0.118	***	0.082	***	0.105	***	0.231
	(2.06)		(3.43)		(3.41)		(3.00)		(2.45)
θ	0.795	***	0.762	***	0.875	***	0.833	***	0.819
	(11.45)		(10.45)		(11.48)		(14.85)		(18.54)
φ	0.140		0.113		0.047		0.121		-0.101
	(1.31)		(1.06)		(0.50)		(1.27)		(-0.86)
LLK	4339.39		4052.17		4205.53		3789.40		3024.76
BIC	-8648.01		-8073.58		-8380.28		-7548.03		-6018.75
Panel B. Inclusion of information arrival (<i>In trading time</i>)									
ω	0.00001	***	0.00010	***	0.00005	***	0.00004	***	0.00003
	(2.65)		(21627.86)		(120.91)		(3.00)		(256.07)
λ	0.070	***	0.063	***	0.044	***	0.076	***	0.278
	(188.47)		(1616.96)		(116.07)		(3.49)		(72.23)
θ	0.863	***	0.829	***	0.889	***	0.878	***	0.841
	(263.17)		(83.69)		(505.14)		(623.88)		(122.52)
φ	0.138	***	0.096	***	0.057	***	0.105	***	-0.031
	(26.01)		(299.35)		(103.99)		(3.51)		(-59.24)
ξ	0.0011	***	0.0009	***	0.0009	***	0.0008	***	0.0004
	(74.24)		(4898.47)		(120.05)		(8.58)		(40.33)
LLK	4618.19		4139.72		4334.99		3878.39		3073.35
BIC	-9197.91		-8240.98		-8631.52		-7718.31		-6108.23

The t-statistics in parentheses are measured with robust standard errors. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

Table 4 shows significant ARCH, GARCH and leverage effects in GJR-GARCH(1,1) with stable distribution. In calendar time (Table 4(A)), all portfolios exhibit pronounced persistence of volatility ($\theta \approx 0.9$). The ARCH effects are significant while lower

⁵ The infinite variance and non-stationary features indicate that a quartile regression method used by previous studies may generate unreliable estimates (see Chan and Zhang, 2009).

($\lambda < 0.1$) than the GARCH effects in all portfolios. Portfolio 1 has the most prominent GARCH effects ($\theta = 0.937$). Compared to the Gaussian assumption (Table 3(A)), the leverage effects (γ) are significant and negative from Portfolio 1 to 4, indicating good news have larger impact on the conditional volatility in calendar time. While in the smallest Portfolio 5, bad news causes greater volatility as the liquidity is low. The constant terms ϖ are all significant, indicating inefficient market. In Table 4(B), the inclusion of information arrival is significant across all portfolios, which is positively associated with the volatility of unexpected price movement (h_t). The impact of volumes on extreme returns varies with the size of cryptos, which aligns with Westerfield (1977) in stock markets. Figure 2 shows similar patterns of conditional volatility and volume. Additionally, in trading time, the tails of innovation's distribution become lighter with higher α . The results show that volumes significantly increase conditional volatility but reduce tail heaviness of crypto returns.

In trading time (Table 4(B) & Fig. 2(B)), the GARCH effect shows the largest decrease ($\theta = 0.425$) in Portfolio 1 compared to that in calendar time (Table 4(A) & Fig. 2(A)), suggesting volume acts as a significant subordinator directing the return generating process of liquid cryptos. For Portfolio 2 and 3, the volatility persistence magnifies. With lower volumes compared to Portfolio 1, medium size cryptos have lower liquidity and news is not quickly reflected by prices, so the information cumulates and volatility spikes decay relatively slowly. For Portfolio 5, the ARCH effect increases after considering information flow, showing stronger influence of short-term volatility spikes when the market is illiquid. According to LeBaron (1992), nontrading and illiquidity could cause serial correlation. With low trading volumes, the behavior of “small cryptos” is similar to “small firms” that may suffer from infrequent trading and thus bias systematic risk estimates (Banz, 1981; Dimson, 1979). In trading time, leverage effects become positive across all portfolios, indicating negative news increase volatility more than positive shocks. Since financial markets are exposed to information shocks about global contagion risk (Arteaga-Garavito et al., 2024), the impacts during COVID-19 are presented in Appendix B.

Table 4. Parameter Estimates of Stable-GJR-GARCH(1,1) Model

	1	2	3	4	5
Panel A. Without information arrival (<i>In calendar time</i>)					
ϖ	0.0002 *** (74.44)	0.0009 *** (41.16)	0.0005 *** (63.38)	0.0004 *** (86.76)	0.0006 *** (111.94)

λ	0.046 ***	0.078 ***	0.062 ***	0.052 ***	0.032 ***
	(272.84)	(148.95)	(171.55)	(188.21)	(190.22)
θ	0.937 ***	0.867 ***	0.901 ***	0.918 ***	0.927 ***
	(2660.46)	(449.76)	(921.36)	(1767.95)	(2187.72)
φ	-0.012 ***	-0.005 ***	-0.010 ***	-0.003 ***	0.003 ***
	(-63.14)	(-7.98)	(-33.38)	(-8.26)	(21.72)
β	-0.012 ***	-0.359 ***	-0.214 ***	-0.087 ***	-0.051 ***
	(-26.22)	(-166.11)	(-118.44)	(-63.88)	(-37.71)
α	1.612 ***	1.752 ***	1.718 ***	1.606 ***	1.532 ***
	(502.1)	(334.76)	(381.45)	(563.5)	(685.16)
LLK	4559.63	4273.91	4088.46	3739.28	3745.71
BIC	-9099.21	-8527.77	-8156.87	-7458.52	-7471.38
Panel B. Inclusion of information arrival (<i>In trading time</i>)					
ϖ	0.0091 ***	0.0005 ***	0.0004 ***	0.0004 ***	0.0006 ***
	(1309.55)	(36.67)	(123.29)	(76.56)	(129.81)
λ	0.003 ***	0.042 ***	0.032 ***	0.049 ***	0.033 ***
	(10.24)	(80.5)	(174.6)	(171.2)	(211.9)
θ	0.425 ***	0.908 ***	0.930 ***	0.911 ***	0.923 ***
	(929.62)	(613.67)	(2292.67)	(1605.15)	(2338.83)
φ	0.169 ***	0.029 ***	0.019 ***	0.015 ***	0.007 ***
	(474.37)	(74.1)	(103.61)	(69.99)	(38.95)
ξ	0.027 ***	0.015 ***	0.012 ***	0.009 ***	0.005 ***
	(2020.19)	(549.03)	(653.4)	(329.11)	(363.57)
β	-0.167 ***	-0.747 ***	-0.564 ***	-0.188 ***	-0.089 ***
	(-78.21)	(-284.21)	(-223.18)	(-132.72)	(-63.27)
α	1.761 ***	1.823 ***	1.809 ***	1.640 ***	1.565 ***
	(318.1)	(301.91)	(283.05)	(493.46)	(637.49)
LLK	4712.47	4366.63	4522.89	4121.60	3767.70
BIC	-9401.55	-8709.87	-9022.39	-8219.82	-7512.02

The t-statistics in parentheses are measured with robust standard errors. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

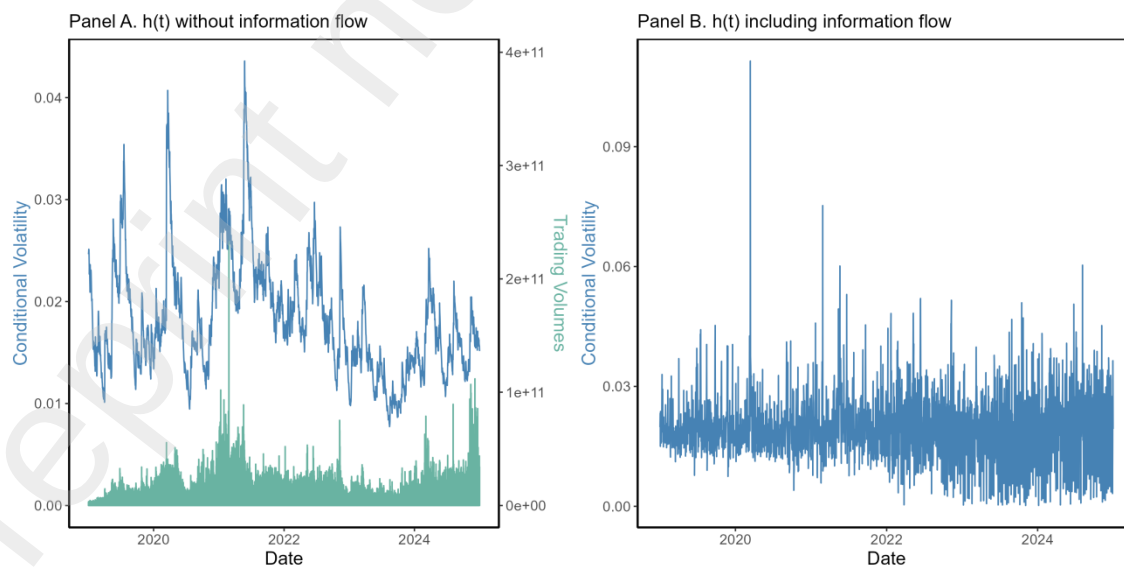


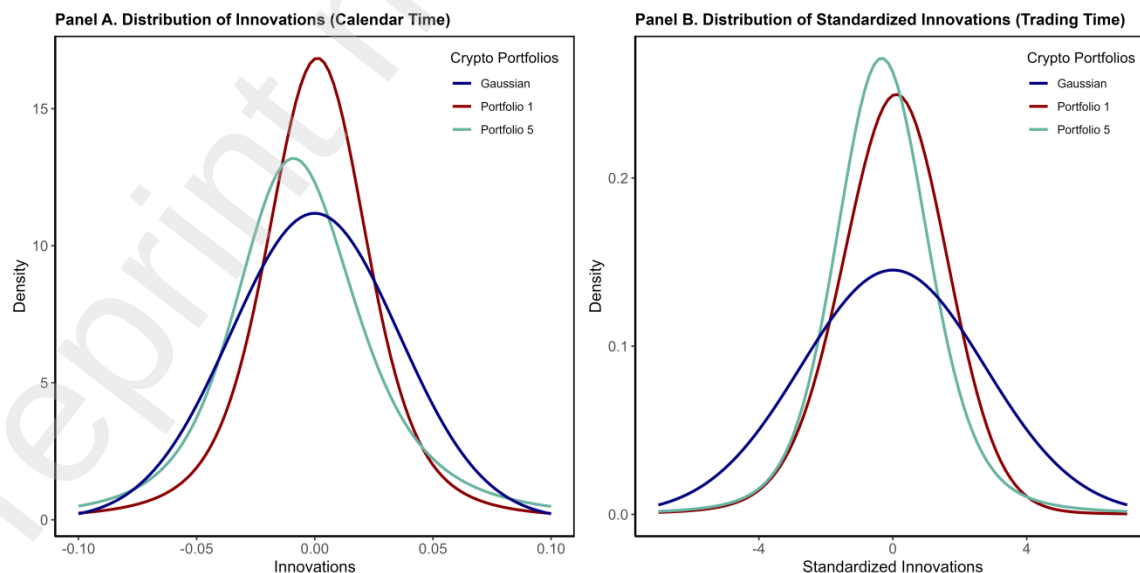
Fig. 2. Conditional Volatility Before and After Including Information Flow (Portfolio 1)

After standardizing information-induced heteroskedasticity, crypto returns and innovations show reduction in tail heaviness. Larger and more liquid cryptos demonstrate lighter tails in *trading time* (Table 4). Portfolios 1 has the lightest tails of standardized innovations with α increasing from 1.543 to 1.872 (Welch's t-test=637.18). However, the characteristic exponents are still significantly less than 2.0. Thus, the log-stable recommended by Mandelbrot and Taylor (1967) is more appropriate than log-normal for describing the returns within both *trading time* and *calendar time*⁶.

Table 4. Stable Distribution Parameter Estimates of Innovations

	1		2		3		4		5	
Panel A: Innovations ε_t										
α	1.543	***	1.631	***	1.609	***	1.460	***	1.383	***
	(-653.15)		(537.84)		(547.92)		(781.99)		(903.74)	
β	-0.047	***	-0.322	***	-0.172	***	0.040	***	0.179	***
	(-32.00)		(-195.91)		(-105.82)		(30.62)		(153.58)	
γ	0.017		0.021		0.019		0.021		0.022	
δ	0.001		0.004		0.002		-0.002		-0.008	
Panel B: Standardized Innovations ε_t/h_t										
α	1.872	***	1.723	***	1.739	***	1.621	***	1.674	***
	(172.61)		(331.87)		(301.66)		(398.86)		(341.66)	
β	-0.544	***	-0.843	***	-0.647	***	-0.196	***	0.115	***
	(-125.29)		(-462.46)		(-263.69)		(-87.4)		(46)	
γ	1.131		0.979		0.985		0.972		1.048	
δ	0.085		0.299		0.204		0.017		-0.320	

The t-statistics are presented in parentheses. To examine whether the distribution is significantly different from normal distribution, the null hypothesis for α is: $\alpha < 2$; to determine whether the distribution is asymmetric, the null hypothesis for β is: $\beta = 0$. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.



assumption is not appropriate because, by definition, the fractal model will not be normal.

Fig. 3. Distributions in Calendar Time and Trading Time

Conclusion

Trading volumes increase cryptocurrency return volatility but reduce distribution tail heaviness, aligning with Fry's (2018) finding that liquidity risks induce heavy tails. It is also consistent with Dyhrberg et al. (2018) that trading induces higher volatility in Bitcoin. The log-stable is more appropriate for describing the information flow within *trading days and calendar days*. Unlike Lamoureux and Lastrapes (1990,1994), this study verifies that the GARCH effect remains significant after including trading volumes as proxy of latent information variables. Our tests are less biased because the information impacts on long-term volatility are captured by the characteristic exponents and thus decreases endogeneity between volatility and volume.

Researchers are recommended to examine the goodness of fit of the “tail risk” in stable Paretian market (Fama, 1965a) and the GARCH-in-Mean model (Engle et al., 1987) for the return generating process of cryptocurrencies. The asset pricing models, in turn, will be helpful for developing economic models to prevent extreme losses, especially during economic crises. The diversification effect will be lower with stable distribution compared to the Gaussian assumption (Fama, 1965b). Potential limitation of this paper is subject to the proxy of information flow because volume could be contaminated by non-informed trading (Brogaard et al., 2014). We examine two alternative proxies of information flow in Appendix C which confirms our findings are robust.

Appendix A: Statistical Relations of Information and Subordinated Distribution

The EMH states that the anticipated change in the price between today and tomorrow cannot be predicted by anything that influenced the past prices. The best forecast of tomorrow's price P_{t+1} is today's price P_t :

$$E(P_{t+1}|P_t, P_{t-1}, P_{t-2}, \dots) = P_t \quad (\text{A.1})$$

EMH only requires market to be a “martingale” or a “fair game”, that means information cannot be used to make arbitrage profits:

$$E(X_{t+1}|\Phi_t) = X_t \quad (\text{A.2})$$

where X_t is a martingale with respect to the information set Φ_t . On the other hand, a stochastic process Y_t is a fair game if it has the following property:

$$E(Y_{t+1}|\Phi_t) = 0 \quad (\text{A.3})$$

The subordinated normal model states that each price change measured over calendar time can be thought of as the sum of a random number, n , of identically distributed random variables, X (Westerfield, 1977). The return of P_t in equation (A.1) can be expressed as:

$$r_t = \mu_t + \sum_{i=1}^{n(t)} e_i \quad (\text{A.4})$$

where $e_i \sim i.i.d. N(0, \sigma^2)$, μ_t denotes forecastable component, e_i are incremental changes. $n(t)$ is the number of new information in period t (within trading days). Because of the randomness of $n(t)$, r_t is not normally distributed. r_t is a subordinated stochastic process, where $r_t - \mu_t$ is subordinated to e_i , and $n(t)$ is the directing process. Bera and Higgins (1993) show that equation (A.4) can be rewritten as:

$$r_t = \mu_t + \gamma n(t)^{1/2} \psi_t \quad (\text{A.5})$$

where ψ_t is standard normal distribution $N(0,1)$. Assuming that the information Φ_t include all current and historical information, r_t is conditional on the information set and the directing process, $n(t)$:

$$r_t | (\Phi_{t-1}, n(t)) \sim N(\mu_t, \gamma^2 n(t)) \quad (\text{A.6})$$

which is a conditional heteroskedastic normal distribution. Since $n(t)$ is not observable and if a random variable condition on $n(t)$ is normal, unconditionally, it must be non-normal as shown by stable distribution. Moreover, the distribution for $r_t | \Phi_{t-1}$, would be conditionally heteroskedastic and non-normal. That is closely related to the GARCH by Bollerslev (1986).

Appendix B: Structural Changes and Conditional Volatility— COVID-19

The parameter estimates of the Stable-GJR-GARCH model during COVID-19 are illustrated in Table B.1. Consistent with our main findings, the impacts of volume on conditional volatility are significant and positive in all portfolios. Also, the tails become lighter after accounting for information flow. The reduction of tail heaviness and the volume effects both increase with the size of cryptos. Moreover, the leverage effect (φ) remains positive in trading time. During COVID-19, the volume's impacts either remain the same or higher in all portfolios compared to that in Table 4. The GARCH effects drop dramatically not only in the largest Portfolio 1 but in Portfolio 2, suggesting information shocks during market turmoil enhance its directing role as subordinator.

Table B.1 Parameter Estimates of Stable-GJR-GARCH(1,1) During COVID

	1	2	3	4	5
Panel A. Without information arrival (<i>In calendar time</i>)					
ω	0.0005 *** (91.25)	0.0015 *** (42.3)	0.0014 *** (68.54)	0.0006 *** (55.62)	0.0004 *** (80.66)
λ	0.049 *** (180.22)	0.084 *** (157.88)	0.079 *** (213.89)	0.051 *** (110.12)	0.036 *** (149.05)
θ	0.912 *** (1735.75)	0.835 *** (351)	0.830 *** (563.54)	0.907 *** (816.17)	0.938 *** (1684.46)
φ	0.000 *** (-6.22)	0.005 *** (5.37)	0.017 *** (26.3)	-0.002 *** (-9.57)	-0.004 *** (-16.36)
β	-0.068 *** (-21.26)	-0.448 *** (-494.76)	-0.216 *** (-84.19)	-0.068 *** (-27.41)	-0.192 *** (-96.36)
α	1.674 *** (318.77)	1.786 *** (852.23)	1.726 *** (279.25)	1.607 *** (425.36)	1.650 *** (358.01)
LLK	2322.70	2173.61	2273.30	2277.19	2051.90
BIC	-4625.35	-4327.17	-4526.55	-4534.33	-4083.76
Panel B. Inclusion of information arrival (<i>In trading time</i>)					
ω	0.0081 *** (482.64)	0.0037 *** (212.04)	0.0012 *** (94.19)	0.0009 *** (73.76)	0.0005 *** (85.5)
λ	0.046 *** (80.86)	0.081 *** (216.39)	0.051 *** (188.32)	0.064 *** (147.2)	0.038 *** (177.79)
θ	0.506 *** (629.5)	0.696 *** (789.35)	0.857 *** (954.89)	0.862 *** (707.14)	0.929 *** (1853.87)
φ	0.128 *** (153.52)	0.102 *** (200.85)	0.050 *** (104.31)	0.015 *** (40.46)	0.003 *** (10.81)
ξ	0.027 *** (1516.57)	0.019 *** (525.1)	0.013 *** (464.36)	0.009 *** (211.44)	0.006 *** (244.67)
β	-0.358 *** (-75.04)	-0.680 *** (-220.16)	-0.601 *** (-161.82)	-0.164 *** (-84.95)	-0.237 *** (-114.11)
α	1.900 *** (138.88)	1.800 *** (257.16)	1.812 *** (205.98)	1.628 *** (390.57)	1.678 *** (332.05)
LLK	2443.99	2213.73	2318.24	2292.81	2070.38
BIC	-4864.59	-4404.07	-4613.09	-4562.22	-4117.38

The t-statistics in parentheses are measured with robust standard errors. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

Appendix C: Robustness Tests –Alternative Information Proxies and Distributions

As robustness tests, we substitute volume with other information proxies, such as the number of transactions (Jones et al., 1994) and the active address (Cong et al., 2021, Rzayev et al., 2025). Table C.1 shows significant and positive impact of information proxies on volatility. Also, tail heaviness reduces with the inclusion of information proxies. The results confirm our main findings.

Table C.1 Parameter Estimates of Stable-GJR-GARCH(1,1) Using Other Information Proxy

	BTC		ETH		LTC		TRX		ADA	
Panel A. Without information arrival										
ω	0.00054 ***		0.00034 ***		0.00081 ***		0.00055 ***		0.00309 ***	
	(16.78)		(45.23)		(41.46)		(147.62)		(168.45)	
λ	0.088 ***		0.070 ***		0.079 ***		0.126 ***		0.160 ***	
	(34.73)		(107.36)		(107.96)		(306.59)		(294.13)	
θ	0.857 ***		0.904 ***		0.873 ***		0.832 ***		0.699 ***	
	(163.88)		(774.19)		(445.36)		(1280.54)		(566.37)	
φ	-0.003 ***		-0.011 ***		-0.014 ***		-0.038 ***		-0.039 ***	
	(-5.54)		(-54.81)		(-32.37)		(-123.76)		(-98.67)	
β	-0.013 **		-0.153 ***		-0.142 ***		-0.023 ***		-0.012 ***	
	(-2.07)		(-78.87)		(-73.39)		(-16.18)		(-14.6)	
α	1.731 ***		1.771 ***		1.738 ***		1.623 ***		1.749 ***	
	(148.48)		(325.28)		(360.59)		(517.34)		(367.33)	
LLK	5133.10		4601.82		4343.58		4491.51		4125.43	
BIC	-10246.16		-9183.60		-8667.12		-8962.98		-8230.81	
Panel B. Inclusion of information arrival (proxied by the change of No. transaction)										
ω	0.00145 ***		0.00052 ***		0.00113 ***		0.00059 ***		0.00232 ***	
	(55.41)		(42.44)		(26.7)		(151.14)		(113.91)	
λ	0.143 ***		0.075 ***		0.087 ***		0.131 ***		0.110 ***	
	(101.17)		(97.26)		(94.32)		(316.76)		(169.01)	
θ	0.735 ***		0.884 ***		0.844 ***		0.825 ***		0.768 ***	
	(228.63)		(586.64)		(228.75)		(1273.82)		(518.48)	
φ	-0.003 ***		0.001 ***		-0.007 ***		-0.039 ***		0.010 ***	
	(-10.23)		(15.85)		(-5.55)		(-129.23)		(32.56)	
ξ	0.008 ***		0.033 ***		0.009 ***		0.005 ***		0.029 ***	
	(194.6)		(830.77)		(145.48)		(85.8)		(608.21)	
β	-0.050 ***		-0.282 ***		-0.161 ***		-0.017 ***		-0.330 ***	
	(-26.17)		(-108.49)		(-73.99)		(-11.18)		(-109.07)	
α	1.769 ***		1.827 ***		1.759 ***		1.623 ***		1.860 ***	
	(278.35)		(243.72)		(298.79)		(524.93)		(222.72)	
LLK	5142.73		4648.08		4355.38		4493.60		4214.86	
BIC	-10262.07		-9272.78		-8687.36		-8963.82		-8406.32	

Panel C. Inclusion of information arrival (proxied by the change of active address)

ϖ	0.00078 *** (62.62)	0.00042 *** (41.84)	0.00095 *** (21.52)	0.00057 *** (156.93)	0.00179 *** (146.06)
λ	0.098 *** (118.19)	0.067 *** (90.36)	0.081 *** (80.36)	0.126 *** (315.17)	0.090 *** (187.72)
θ	0.829 *** (482.47)	0.903 *** (659.76)	0.859 *** (208.27)	0.830 *** (1324.32)	0.815 *** (888.98)
φ	0.003 *** (18.33)	-0.007 *** (-40.27)	-0.006 *** (-3.57)	-0.036 *** (-102.47)	0.011 *** (39.07)
ξ	0.021 *** (383.42)	0.012 *** (138.06)	0.012 *** (153.63)	0.007 *** (117.62)	0.029 *** (794)
β	-0.067 *** (-29.59)	-0.198 *** (-90.92)	-0.174 *** (-84.95)	-0.015 *** (-2.32)	-0.382 *** (-105.89)
α	1.793 *** (227.71)	1.789 *** (287.35)	1.759 *** (297.62)	1.629 *** (510.42)	1.890 *** (171.38)
LLK	5182.21	4607.39	4360.54	4496.07	4231.73
BIC	-10341.03	-9191.39	-8697.70	-8968.76	-8440.06

Sample period: 2019.1.1-2025.1.1. The data are obtained from IntoTheBlock.com, which provides the on-chain data for Bitcoin (BTC), Ethereum (ETH), Litecoin (LTC), TRON (TRX), ADA. The t-statistics in parentheses are measured with robust standard errors. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

As another robustness test, we model the crypto portfolios' innovations using other heavy-tailed distributions. The results in Table C.2 reaffirm our main conclusion. Stable distribution satisfies the Generalized Central Limit Theorem that the other distributions could not meet. The Generalized Central Limit Theorem is important because it is the correct mathematical tool for modeling the sum of financial returns. Stable distribution describes the observed fat tails and extreme events that the classic Gaussian model fails to capture and includes Gaussian as a special case ($\alpha=2$). It provides a more robust foundation for asset pricing, portfolio theory, and, most critically, risk management, where accurately gauging the probability of catastrophic loss is paramount. It shifts the perspective from seeing extreme price movements as outliers to seeing them as inherent properties of the market's mathematical structure. Therefore, stable distribution is more appropriate.

Table C.2 Parameter Estimates of GJR-GARCH(1,1) Under Other Distributions

	1	3	5		1	3	5	
Panel A. Skewed Student-t Distribution								
A.1 Without information arrival				A.2 Inclusion of information arrival				
ϖ	0.00002 (0.8)	0.00005 (1.17)	0.00024 (2.01)	**	0.00004 (216.41)	0.00004 (18.75)	0.00033 (6.2)	***
λ	0.080 (3.08)	0.135 (3.59)	0.050 (2.36)	**	0.092 (101.77)	0.041 (5.54)	0.074 (2.27)	**
θ	0.925	0.872	0.881	***	0.842	0.917	0.893	***

	(18.53)		(14.27)		(22.19)		(1331.42)		(7.45)		(559.43)	
φ	-0.012		-0.030		0.143	**	0.036	***	0.027	***	0.310	***
	(-0.29)		(-0.66)		(2.43)		(96.75)		(11.03)		(2.92)	
ξ							0.0015	***	0.0008	***	0.0021	***
							(123.2)		(7.99)		(4.73)	
β	0.943	***	0.838	***	0.944	***	0.920	***	0.809	***	0.934	***
	(56.13)		(40.97)		(53.78)		(41.59)		(22.67)		(54.5)	
α	3.106	***	3.597	***	2.468	***	5.213	***	4.610	***	2.247	***
	(13.52)		(11.04)		(20.83)		(10.09)		(0.50)		(41.68)	
LLK	4567.17		4432.94		3751.21		4752.41		4521.26		3783.03	
BIC	-9088.18		-8819.73		-7456.25		-9450.97		-8988.68		-7512.21	

Panel B. Normal Inverse Gaussian Distribution

B.1 Without information arrival

B.2 Inclusion of information arrival

ϖ	0.00003		0.00004		0.00020	**	0.00002	***	0.00003	***	0.00017	***
	(1.11)		(1.28)		(2.39)		(9.61)		(58.9)		(3.52)	
λ	0.071	***	0.123	***	0.057	**	0.083	***	0.020	***	0.047	***
	(3.33)		(3.87)		(2.21)		(16.87)		(31.48)		(2.61)	
θ	0.914	***	0.876	***	0.866	***	0.871	***	0.945	***	0.886	***
	(21.33)		(16.92)		(27.99)		(16.78)		(49.93)		(154.34)	
φ	0.000		-0.027		0.159	**	0.049	***	0.014	***	0.145	***
	(0.01)		(-0.68)		(2.28)		(5.35)		(28.66)		(2.85)	
ξ							0.001	***	0.001	***	0.001	***
							(33.76)		(50.81)		(7.78)	
β	-0.116	***	-0.285	***	-0.118	***	-0.143	***	-0.332	***	-0.115	***
	(-3.78)		(-7.89)		(-2.95)		(-3.85)		(-7.19)		(-2.86)	
α	0.518	***	0.737	***	0.203	***	1.443		1.013		0.215	***
	(6.61)		(5.67)		(3.81)		(1.33)		(1.29)		(3.91)	
LLK	4572.31		4432.34		3737.81		4749.08		4503.02		3759.73	
BIC	-9098.45		-8818.53		-7429.46		-9444.30		-8952.20		-7465.61	

Panel C. Generalized Error Distribution

C.1 Without information arrival

C.2 Inclusion of information arrival

ϖ	0.00004		0.00004		0.00013	***	0.00000		0.00006	***	0.00012	***
	(1.29)		(1.23)		(2.73)		(0.24)		(16.9)		(3.83)	
λ	0.066	***	0.094	***	0.066	**	0.041	***	0.058	***	0.055	***
	(3.29)		(4.04)		(2.53)		(30.49)		(30.52)		(3.04)	
θ	0.896	***	0.883	***	0.862	***	0.940	***	0.872	***	0.878	***
	(17.89)		(18.31)		(30.65)		(525.46)		(29.75)		(343.11)	
φ	0.019		0.000		0.115	**	0.018	***	0.052	***	0.109	**
	(0.46)		(0.00)		(2.09)		(48.5)		(128.79)		(2.55)	
ξ							0.001	***	0.001	***	0.001	***
							(303.09)		(43.65)		(8.68)	
α	0.961	***	1.092	***	0.778	***	1.138	***	1.196	***	0.788	***
	(20.24)		(17.15)		(16.36)		(11.93)		(12.66)		(15.85)	
	4562.83		4383.32		3692.50		4689.28		4458.20		3712.65	
	-9087.20		-8728.18		-7346.53		-9332.40		-8870.25		-7379.14	

The t-statistics in parentheses are measured with robust standard errors. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively. α and β denote the shape and skewness parameters in respective distribution.

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Figure Captions

Fig. 1. Price, Volatility and Volumes of Bitcoin

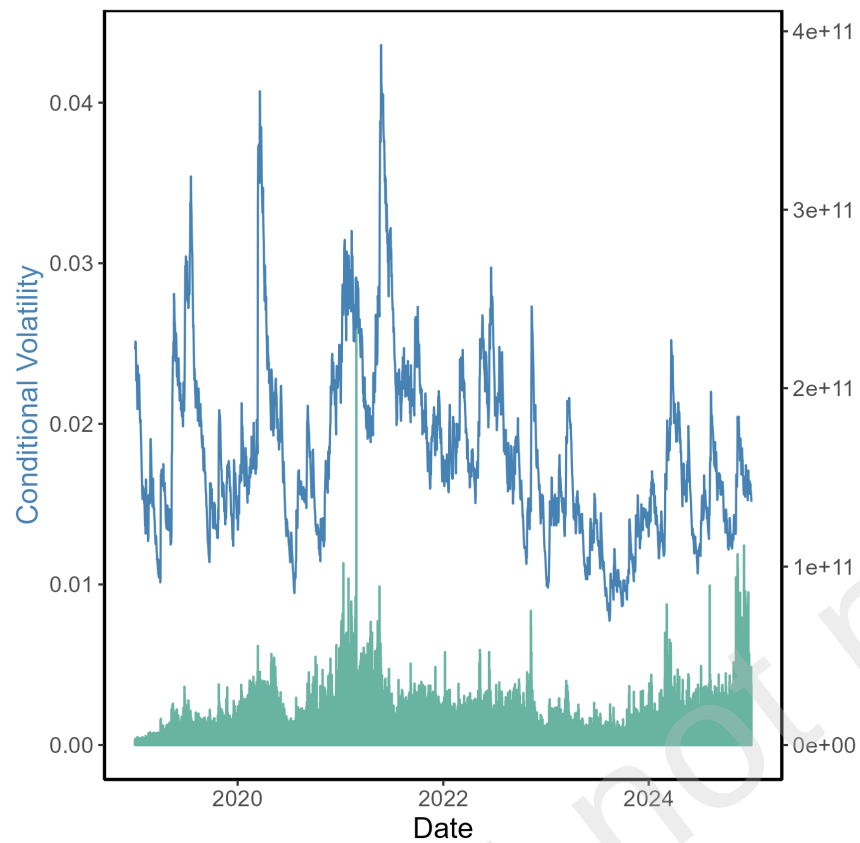
Fig. 2. Conditional Volatility Before and After Including Information Flow (Portfolio 1)

Fig. 3. Distributions in Calendar Time and Trading Time

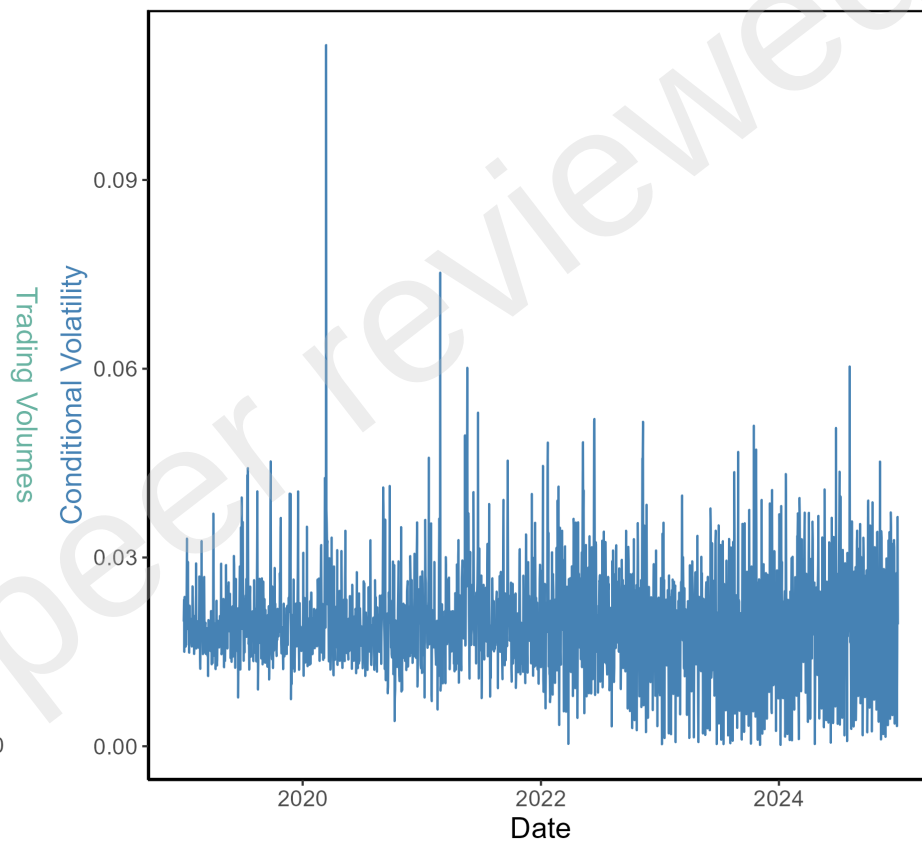
Price, Volatility and Volumes of Bitcoin (1.1.2015~1.1.2025)



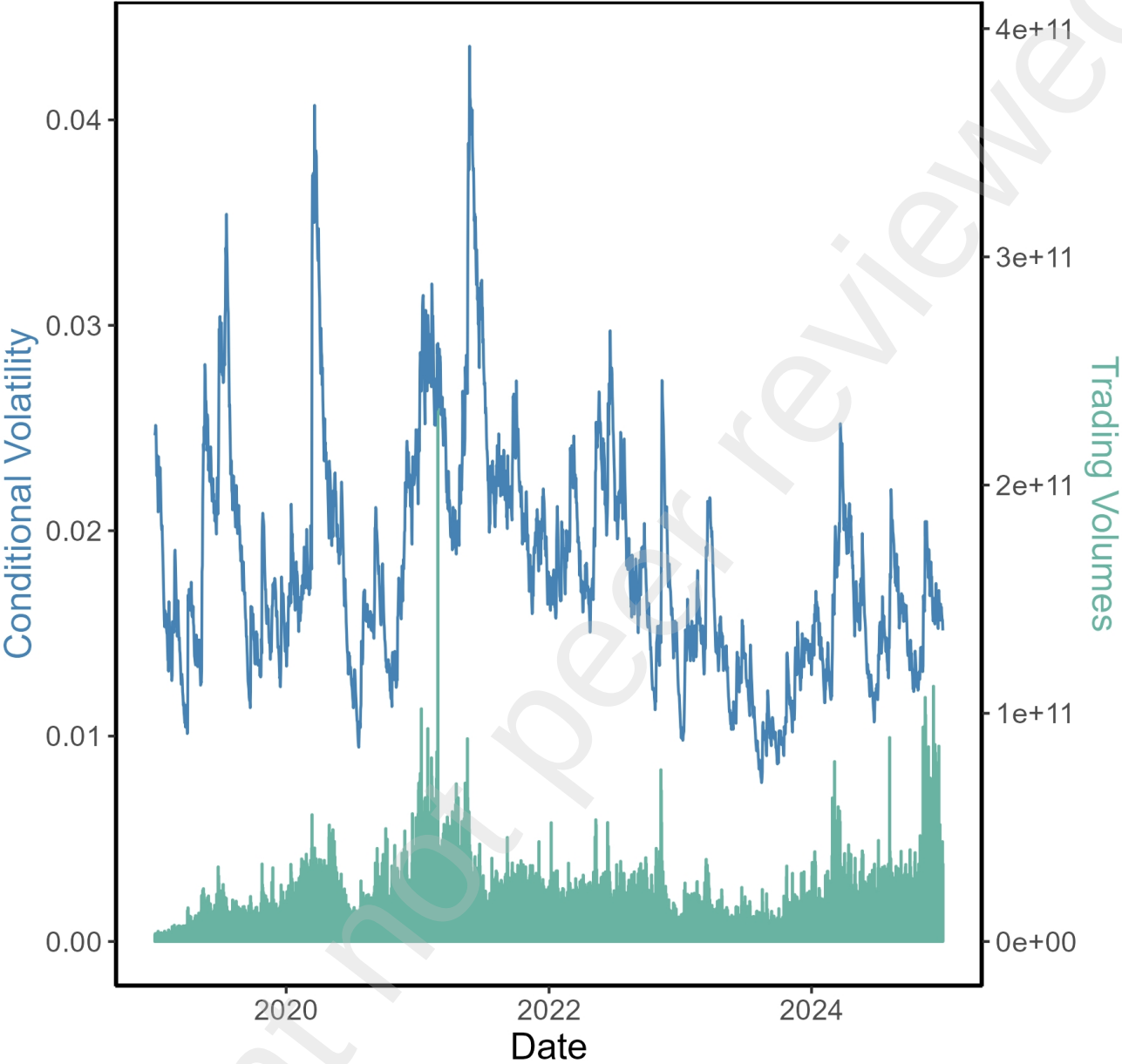
Panel A. $h(t)$ without information flow



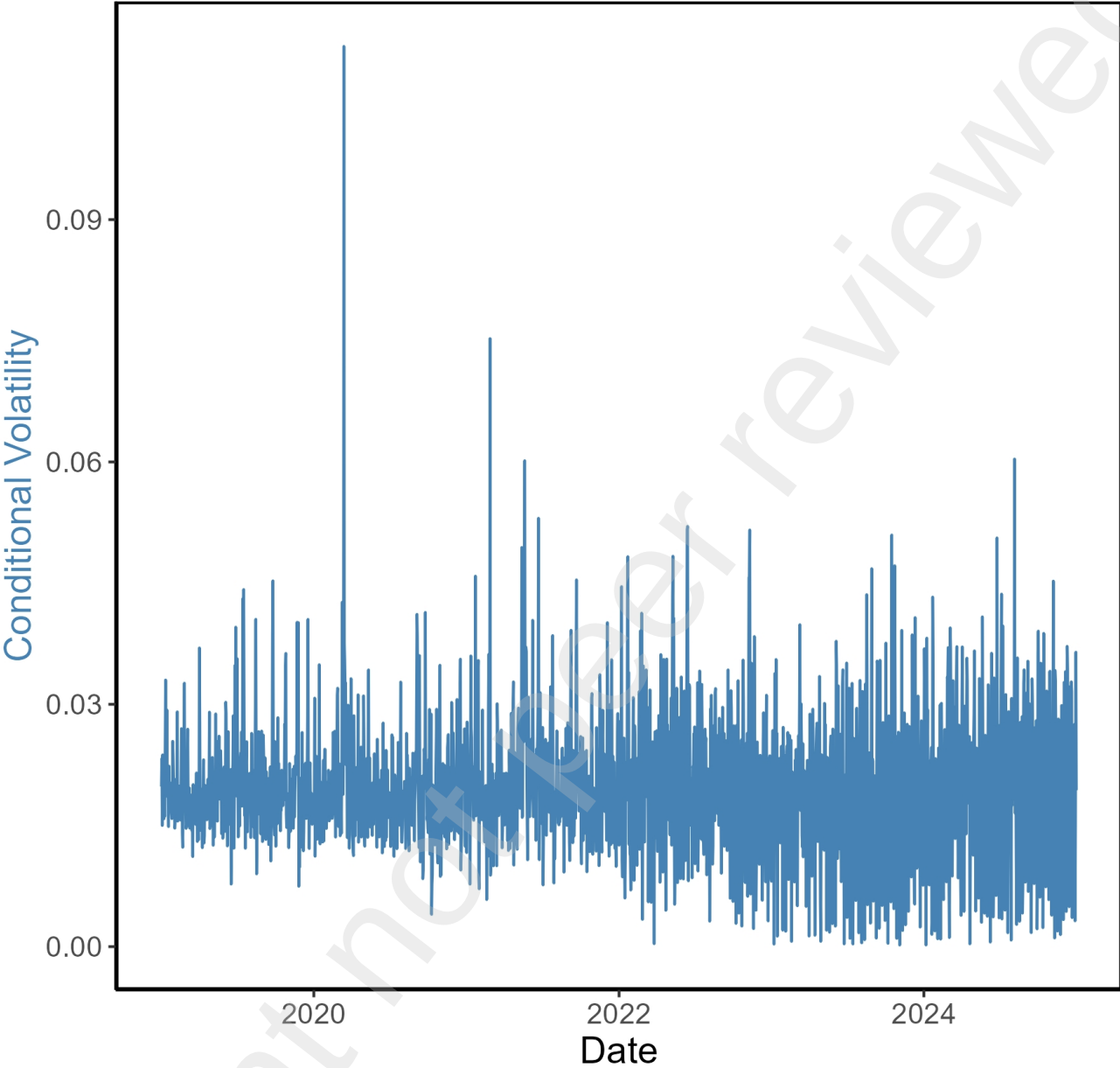
Panel B. $h(t)$ including information flow



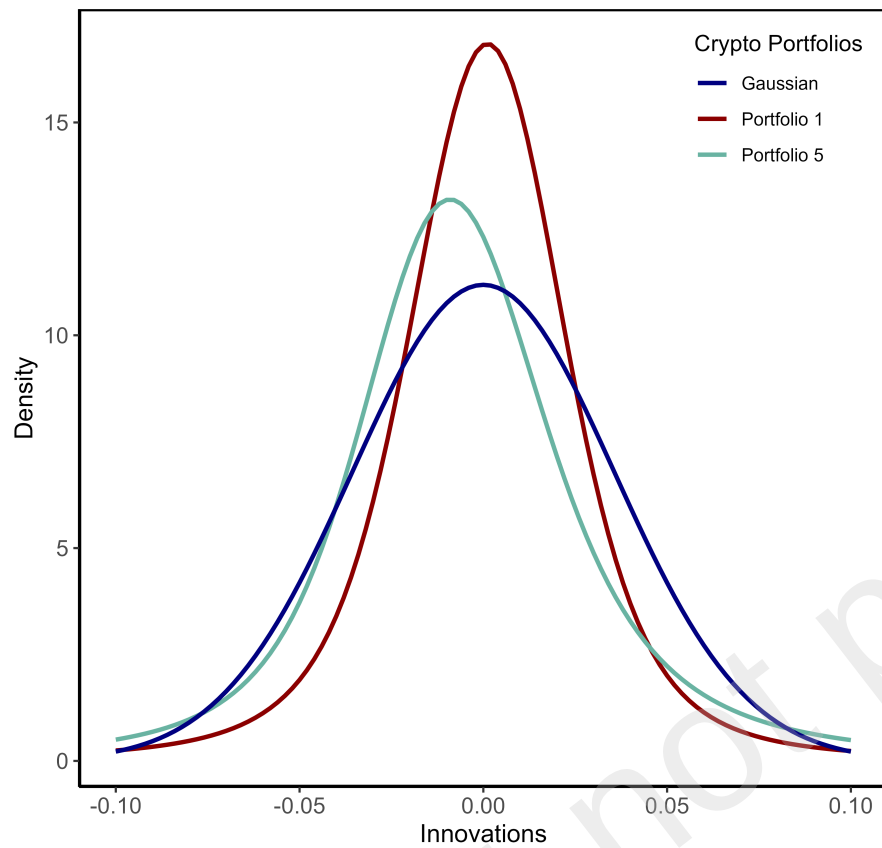
Panel A. $h(t)$ without information flow



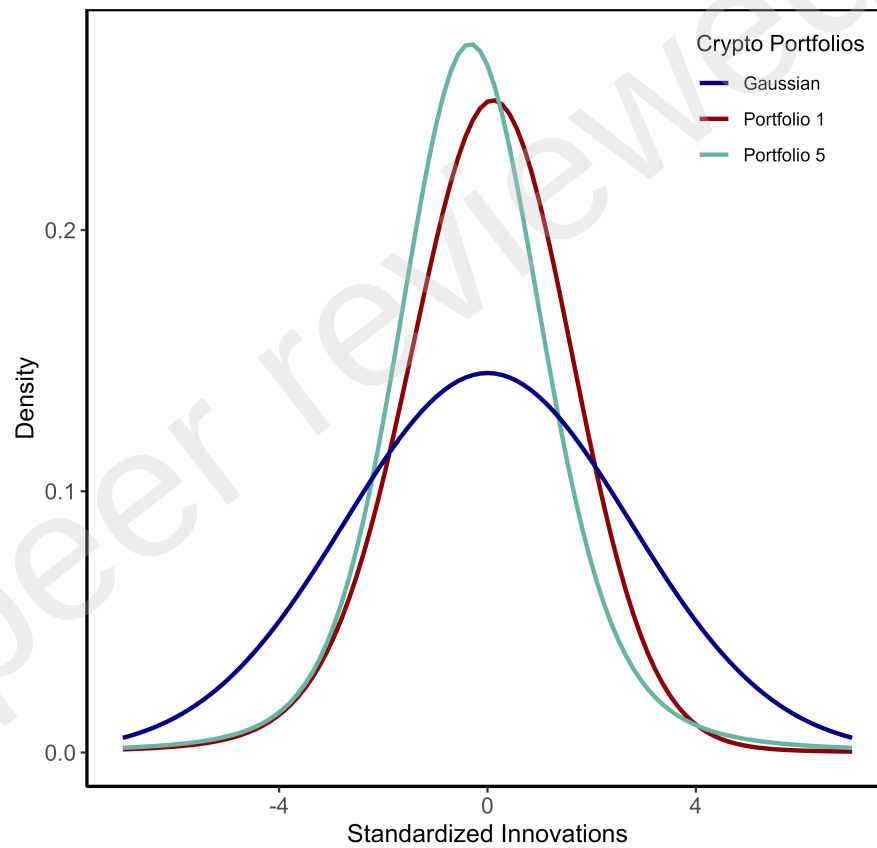
Panel B. $h(t)$ including information flow



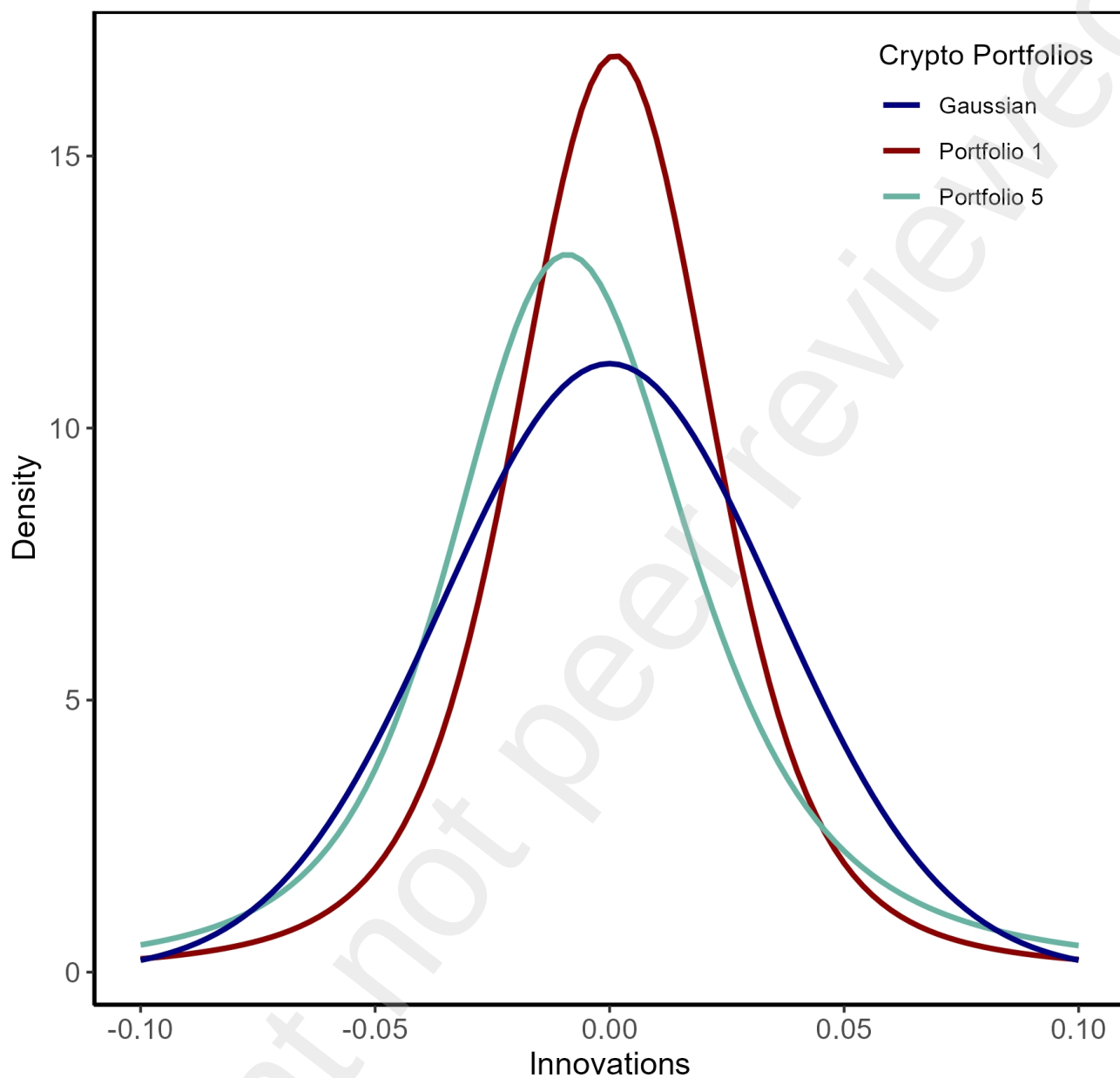
Panel A. Distribution of Innovations (Calendar Time)



Panel B. Distribution of Standardized Innovations (Trading Time)



Panel A. Distribution of Innovations (Calendar Time)



Panel B. Distribution of Standardized Innovations (Trading Time)

